

Region Proposal Methods in Object Detection

What Is a Region Proposal?

A region proposal is a candidate bounding box that is likely to contain an object of interest. Rather than classifying every possible window in an image, region proposal methods narrow the search space to a manageable set of promising candidates.

In a two-stage detector, these proposals are passed to a second stage for precise classification and box refinement. The quality of proposals directly limits detection recall: any object missed at the proposal stage cannot be recovered later.

Selective Search

Selective Search was the dominant proposal method before deep learning approaches. It starts with an over-segmentation of the image and then greedily merges regions based on color, texture, size, and fill similarity.

Selective Search typically generates around 2000 proposals per image. Although it is not learned from data, it achieves high recall for objects of various sizes and aspect ratios. Its main drawback is speed: it runs on the CPU and takes about 2 seconds per image.

Region Proposal Networks (RPN)

Faster R-CNN introduced the Region Proposal Network, a small fully convolutional network that slides over the backbone feature map and predicts objectness scores and box offsets for a set of anchor boxes at each location.

The RPN shares convolutional features with the detection head, so proposals are generated almost for free in terms of computation. A typical RPN produces hundreds of thousands of raw proposals, which are then filtered by NMS to a few hundred high-quality candidates.

Training the RPN involves two losses: a binary classification loss (object vs. background for each anchor) and a smooth L1 regression loss for the box offsets of positive anchors.

Anchor Design in RPNs

Anchors are pre-defined reference boxes of different scales and aspect ratios centered at each feature map location. Common choices are three scales (e.g., 128, 256, 512 pixels) and three aspect ratios (1:1, 1:2, 2:1), giving 9 anchors per location.

An anchor is labeled positive if its IoU with any ground-truth box is above 0.7, negative if below 0.3, and ignored otherwise. The regression target for a positive anchor is the offset needed to align the anchor with the ground-truth box.

Non-Maximum Suppression (NMS)

Both RPNs and detection heads produce many overlapping predictions for the same object. Non-Maximum Suppression (NMS) resolves this by iteratively selecting the highest-scoring box and removing all other boxes that overlap with it above an IoU threshold (typically 0.5-0.7).

Soft-NMS is a variant that decays the scores of overlapping boxes rather than removing them, improving recall for crowded scenes where true objects legitimately overlap.

Feature Pyramid Networks (FPN)

Objects appear at different scales in images. Feature Pyramid Networks (FPN) address this by building a multi-scale feature hierarchy. A top-down pathway with lateral connections fuses high-level semantic features with high-resolution low-level features.

RPNs are attached at every level of the FPN, so proposals are generated at multiple scales without the computational cost of processing the image at multiple resolutions. FPN dramatically improves detection of small objects.

Proposal-Free Approaches

Modern one-stage detectors and transformer-based detectors (DETR, DINO) do not use an explicit proposal stage. Instead, detection is formulated as a direct set prediction or dense prediction problem, collapsing the two-stage pipeline into one.

While proposal-free methods are architecturally simpler, understanding region proposal networks remains important for interpreting a large body of literature and for applications where two-stage accuracy is required.

